VYORA

Backend Engineering Blueprint & Complete API Catalog

Technical Architecture & API Reference
Framework: NestJS 11 Modular Monolith
Total Implemented Endpoints: **49 Endpoints**

1. Technical Stack & Engineering Rationale

The Vyora backend is engineered as a decoupled, multi-tenant modular monolith designed for horizontal scaling, absolute data integrity, and strict separation of business concerns:

Technology / Layer	Selected Component	Engineering Rationale & Architectural Purpose
Core Framework	**NestJS 11** (Express adapter)	Provides enterprise architectural conventions: dependency injection, declarative routing, guards, interceptors, and modular domain encapsulation without microservices overhead.
Database & ORM	**MongoDB Atlas + Mongoose 8**	Document-based persistence perfectly mirrors hierarchical financial documents (invoices with embedded line items). Multi-tenancy enforced through indexed `businessId` scoping.
Runtime & Language	**Node.js 20+ / TypeScript 5.8**	Strict typing across all services, DTOs, and schemas. Zero implicit any. Compiles cleanly with `tsc`.
Authentication	**JWT + Passport + bcrypt**	Stateless JSON Web Tokens with 7-day expiration. Passwords securely hashed using bcrypt (salt rounds = 10). User memberships auto-link to business tenants.
Financial Mathematics	**Integer Minor Units (Paisa)**	**Rule of Zero Float:** Every financial value (prices, discounts, taxes, grand totals) is stored and computed in minor units (1 INR = 100 Paisa), eliminating floating-point rounding bugs.
Validation Engine	**Zod + `@vyora/validation`**	Shared, strict runtime validation schemas. Guarantees that payloads rejected by the frontend are also rejected by the backend with deterministic 400 error envelopes.
Security & Rate Limits	**Sliding-Window Memory Throttler**	In-memory rate limiter with periodic cleanup: 25 req/min on `/auth/*` (brute-force protection) and 200 req/min general. Automatic security headers (no-sniff, frame-options).
API Documentation	**Swagger / OpenAPI 3.0**	Auto-generated interactive API documentation available at `/api/v1/docs` with Bearer auth support.

2. Backend Repository & Folder Hierarchy

The backend follows a domain-driven modular structure inside `C:\Users\sande\vyora\vyora-backend`:

```
vyora-backend/ ├── src/ │ ├── common/ # Cross-cutting framework infrastructure │ │ ├── guards/ # JwtAuthGuard, TenantGuard, RolesGuard, PlanGuard │ │ ├── decorators/ # @CurrentUser, @CurrentTenant, @Roles, @PlanRequired │ │ ├── filters/ # AllExceptionsFilter (standardized error envelopes) │ │ └── middleware/ # RateLimitMiddleware, TenantHeaderMiddleware │ ├── modules/ # Domain-driven feature modules (13 core domains) │ │ ├── auth/ # Registration, login, JWT strategy, session verification │ │ ├── business/ # Organization profile, GST settings, staff RBAC membership │ │ ├── products/ # Item catalog, SKU, tax slabs, units, pricing │ │ ├── parties/ # Customers and suppliers, credit tracking, party balances │ │ ├── invoices/ # Sales billing, server tax math, numbering, PDF data │ │ ├── purchases/ # Supplier bills, line items, accounts payable │ │ ├── payments/ # Payment vouchers (Receipt & Payment), invoice allocation │ │ ├── inventory/ # Stock movements ledger, manual adjustments with reasons │ │ ├── expenses/ # Operating expenses, expense categories, payment modes │ │ ├── dashboard/ # Real-time KPIs, sales volume, receivables, low-stock │ │ ├── reports/ # Sales/purchase registers, stock summary, customer dues │ │ ├── audit/ # Financial and administrative audit trails │ │ └── health/ # Health check endpoint for uptime and monitoring │ ├── seeder/ # Automated bootstrap seeder (creates demo tenant & sample data) │ ├── app.module.ts # Root application module coordinating database and routes │ └── main.ts # Application entry point (port 4000, global prefix /api/v1)
```

3. Non-Negotiable Backend Architectural Invariants

- **Multi-Tenancy Isolation (`x-business-id`):** Every mutation and query is scoped to the tenant. `TenantGuard` extracts `x-business-id` from headers, verifies that the requesting JWT user is an active member of that business, and binds the verified ID. Requests with invalid or un-joined business IDs are rejected with `403 Forbidden`.
- **Authoritative Calculation Engine:** Frontend previews totals for user convenience, but the server calculates subtotals, CGST/SGST/IGST, rounding, and grand totals upon submission. Client-calculated totals are never trusted.

- **Immutability of Finalized Documents:** Finalized invoices and purchase bills cannot be directly updated or overwritten. Any correction requires a formal cancellation flow with recorded reason and stock reversal.
- **Traceable Inventory Ledger:** Direct stock overwrites are strictly forbidden. Stock level is altered only via verified movements: Sales deduction (`SALE`), Purchase addition (`PURCHASE`), or Manual Adjustment (`ADJUSTMENT`) with mandatory reason.
- **Idempotency Guarantees:** Invoices and payment vouchers validate uniqueness constraints to ensure network retries cannot double-record bills or payments.

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Complete API Directory • Modules 1 through 5 (15 Endpoints)

API Reference: Section 1
Health, Auth, Business, Staff & Dashboard
Global Prefix: `/api/v1`

4. Core API Directory (15 of 49 Endpoints)

Module 1: Health & Diagnostics (1 Endpoint)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
GET	`/api/v1/health`	Public (None)	None	Returns server uptime, database connectivity status, and release version timestamp for health checks.

Module 2: Authentication & Sessions (4 Endpoints)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
POST	`/api/v1/auth/register`	Public (None)	`{ name, email, password, businessName, phone? }`	Hashes password with bcrypt, creates User document, initializes default Business tenant, assigns user as `OWNER`, and issues JWT token.
POST	`/api/v1/auth/login`	Public (None)	`{ email, password }`	Validates credentials against bcrypt hash, fetches active business memberships, generates signed JWT access token.
GET	`/api/v1/auth/me`	JWTAUTH	Bearer Token header	Returns authenticated user identity, assigned role, active business ID, and user profile data.
POST	`/api/v1/auth/logout`	JWTAUTH	Bearer Token header	Logs logout audit event and clears server-side session references if applicable.

Module 3: Business Profile & Settings (4 Endpoints)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
GET	`/api/v1/business/current`	JWT + TENANT	`x-business-id` header	Fetches current business profile, GSTIN, legal address, invoice prefix, numbering sequence, and active subscription plan.
PUT	`/api/v1/business/current`	OWNER/ADMIN	`BusinessProfileInput` (Zod parsed)	Updates business name, address, state code, GST profile, terms & conditions, and invoice numbering format.
PUT	`/api/v1/business/plan`	OWNER/ADMIN	`{ plan: 'starter' | 'growth' | 'enterprise' }`	Updates business subscription tier. Emits audit log event. Enforces plan feature gating immediately.
PUT	`/api/v1/business/permissions`	OWNER/ADMIN	`Record<string, boolean>`	Updates granular module access permissions for the staff role within this business tenant.

Module 4: Staff & RBAC Management (5 Endpoints)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
GET	`/api/v1/business/staff`	JWT + TENANT	`x-business-id` header	Lists all team members, assigned roles (`OWNER`, `ADMIN`, `STAFF`), active status, and join date.

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
POST	/api/v1/business/staff	OWNER/ADMIN	{ name, email, role }	Invites or creates a staff member, links business membership, assigns initial permissions.
PATCH	/api/v1/business/staff/:id/role	OWNER/ADMIN	{ role: UserRole }	Switches member role. **Primary Owner Immutability:** Blocks attempts to demote primary owner with 403 Forbidden.
PATCH	/api/v1/business/staff/:id/status	OWNER/ADMIN	{ isActive: boolean }	Enables or disables member access. Deactivated staff tokens are rejected by TenantGuard.
DELETE	/api/v1/business/staff/:id	OWNER/ADMIN	Path parameter :id	Removes staff membership from this business tenant. Prevents primary owner self-deletion.

Module 5: Dashboard & Analytics (1 Endpoint)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
GET	/api/v1/dashboard/metrics	JWT + TENANT	Query: timeRange? (30d, 7d, today)	Aggregates gross sales, total invoices, pending receivables, stock valuation, low-stock alerts, and 5 recent invoices.

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Complete API Directory • Modules 6 through 9 (19 Endpoints)

API Reference: Section 2
Products, Parties, Invoices & Purchases
Global Prefix: `/api/v1`

5. Core Domain APIs (19 of 49 Endpoints)

Module 6: Products & Catalog (5 Endpoints)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
POST	`/api/v1/products`	OWNER/ADMIN	`CreateProductDto` (Zod parsed)	Creates product in catalog. Stores prices in paisa. Validates unique SKU per tenant. Records opening stock movement if > 0.
GET	`/api/v1/products`	JWT + TENANT	Query: `search, stockFilter, taxFilter, unit, page, limit`	Returns paginated product list with current stock levels, low-stock flags, tax rates, and pricing in minor units.
GET	`/api/v1/products/:id`	JWT + TENANT	Path parameter `:id`	Fetches product details, complete stock movement history, and pricing specifications.
PUT	`/api/v1/products/:id`	OWNER/ADMIN	`UpdateProductDto`	Updates name, SKU, tax category, selling/purchase prices, reorder point. Does not mutate stock directly (use adjustments).
DELETE	`/api/v1/products/:id`	OWNER/ADMIN	Path parameter `:id`	Deletes product if no active finalized invoices reference it. Otherwise marks deleted or returns validation barrier.

Module 7: Parties Master (Customers & Suppliers) (5 Endpoints)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
POST	`/api/v1/parties`	OWNER/ADMIN	`CreatePartyDto` (Zod parsed)	Creates Customer or Supplier record. Saves contact, state code, GSTIN, and opening balance (stored in paisa).
GET	`/api/v1/parties`	JWT + TENANT	Query: `type, search, balanceFilter, state, page, limit`	Returns paginated parties with live net balance (receivable or payable) and contact details.
GET	`/api/v1/parties/:id`	JWT + TENANT	Path parameter `:id`	Returns party profile, full transaction history (invoices, payments, purchases), and current balance.
PUT	`/api/v1/parties/:id`	OWNER/ADMIN	`UpdatePartyDto`	Updates name, phone, email, address, state code, credit limit, and GSTIN.
DELETE	`/api/v1/parties/:id`	OWNER/ADMIN	Path parameter `:id`	Deletes party if no active balance or invoice references exist. Enforces referential integrity.

Module 8: Sales Invoices & Billing (6 Endpoints)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
POST	`/api/v1/invoices/preview`	JWT + TENANT	`InvoiceDraftInput`	Authoritative server preview: calculates item subtotal, itemized CGST/SGST/IGST in paisa, and grand total without persisting.
POST	`/api/v1/invoices`	OWNER/ADMIN	`CreateInvoiceDto`	Creates invoice in `DRAFT` status. Server calculates totals in minor

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
				units. Assigns draft reference. Stock NOT deducted yet.
GET	/api/v1/invoices	JWT + TENANT	Query: status, search, timeFilter, page, limit	Returns paginated invoices with customer names, dates, grand totals, due amounts, and status badges.
GET	/api/v1/invoices/:id	JWT + TENANT	Path parameter :id	Returns full invoice document with snapshot line items, tax breakdown, customer address, payment history, and business info for PDF print.
POST	/api/v1/invoices/:id/finalize	OWNER/ADMIN	Path parameter :id	**Finalization Engine:** Validates stock availability, assigns permanent sequential invoice number (e.g. INV-2026-0001), deducts inventory traceably, updates customer receivable balance, transitions state to UNPAID. Document becomes immutable.
POST	/api/v1/invoices/:id/cancel	OWNER/ADMIN	{ reason: string }	**Cancellation Engine:** Validates cancellation reason, restores deducted stock back to inventory via reversal movement, reverses customer balance, marks status CANCELLED, records audit event.

Module 9: Purchases & Supplier Bills (3 Endpoints)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
POST	/api/v1/purchases	OWNER/ADMIN	CreatePurchaseDto	Records supplier bill. Calculates tax in paisa, immediately increases physical inventory for each line item, updates supplier payable balance, generates purchase record.
GET	/api/v1/purchases	JWT + TENANT	Query: supplierId, search, timeFilter, page, limit	Returns paginated purchase bills with supplier name, bill number, bill date, total amount, and line count.
GET	/api/v1/purchases/:id	JWT + TENANT	Path parameter :id	Fetches complete purchase bill details with itemized received goods, tax rates, and supplier details.

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Complete API Directory • Modules 10 through 14 (15 Endpoints)

API Reference: Section 3
Payments, Inventory, Expenses, Reports & Audit
Total Endpoints Documented: **49/49**

6. Financial Ledger & Operational APIs (15 of 49 Endpoints)

Module 10: Payments & Vouchers (3 Endpoints)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
POST	`/api/v1/payments`	OWNER/ADMIN	`CreatePaymentDto`	**Payment Allocation Engine:** Records voucher (`RECEIPT` from customer or `PAYMENT` to vendor). Reconciles against invoice `dueMinor`, updates invoice state (`PARTIAL` or `PAID`), adjusts party ledger balance. Enforces unique idempotency key to prevent double posting.
GET	`/api/v1/payments`	JWT + TENANT	Query: `type, method, search, page, limit`	Returns paginated voucher entries with voucher number, party name, method (Cash, Bank, UPI, Cheque), reference/UTR, and amount.
GET	`/api/v1/payments/:id`	JWT + TENANT	Path parameter `:id`	Fetches payment receipt voucher with allocated invoice numbers and payment verification timestamps.

Module 11: Inventory & Stock Adjustments (2 Endpoints)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
GET	`/api/v1/inventory/movements`	JWT + TENANT	Query: `productId, movementType, page, limit`	Returns complete immutable inventory movement audit trail: previous stock, change amount, new stock, reference document ID, actor.
POST	`/api/v1/inventory/adjustments`	OWNER/ADMIN	`{ productId, quantityDelta, reason, notes? }`	Adjusts physical stock. Validates reason enum (`DAMAGED`, `RECOUNT`, `RETURN`, `THEFT`). Updates current stock on product document and records movement audit entry.

Module 12: Expenses Management (4 Endpoints)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
POST	`/api/v1/expenses`	OWNER/ADMIN	`CreateExpenseDto`	Records operational expense (rent, electricity, salary, courier). Saves amount in minor units, payment mode, and optional receipt note.
GET	`/api/v1/expenses`	JWT + TENANT	Query: `category, startDate, endDate, page, limit`	Returns paginated expenses filtered by date range and category.
GET	`/api/v1/expenses/categories`	JWT + TENANT	None	Returns list of active expense categories and aggregate expense sum per category for this tenant.
DELETE	`/api/v1/expenses/:id`	OWNER/ADMIN	Path parameter `:id`	Deletes expense record. Emits audit trail event.

Module 13: Business Reports (5 Endpoints)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
GET	`/api/v1/reports/sales-register`	JWT + TENANT	Query: `startDate, endDate, status`	Detailed sales register: itemized tax breakdown (CGST, SGST, IGST), customer names, invoices, taxable amounts, and gross totals.
GET	`/api/v1/reports/purchase-register`	JWT + TENANT	Query: `startDate, endDate, supplierId`	Purchase bill register with supplier tax numbers, invoice numbers, input tax credit totals, and bill sums.
GET	`/api/v1/reports/stock-summary`	JWT + TENANT	None	Current stock inventory valuation: computes total stock value (quantity × purchase price in paisa) across catalog.
GET	`/api/v1/reports/customer-dues`	JWT + TENANT	None	Outstanding receivables ledger: lists all customers with pending unpaid balance and days overdue.
GET	`/api/v1/reports/expense-summary`	JWT + TENANT	Query: `startDate, endDate`	Comprehensive expense summary grouped by category and payment method with total operating outlay.

Module 14: Audit Trail & Compliance (1 Endpoint)

Method	Endpoint Route	Guards & Access	Payload / Parameters	Business Logic & Side Effects
GET	`/api/v1/audit/logs`	OWNER/ADMIN	Query: `entityType, action, page, limit`	Returns chronological audit history of important financial actions: invoice finalizations, cancellations, stock adjustments, and staff role edits.

> **API Directory Summary:** All 49 endpoints are strictly tenant-isolated, enforce integer minor unit calculations for currency, execute inside NestJS domain services, and are verified by unit/integration tests and OpenAPI contracts.

7. Quality Checks: The Five Layers

To eliminate regressions, guarantee strict typing, maintain uniform style, and prevent broken behavior from ever reaching production, Vyora enforces five defensive quality layers:

Layer	Tool	What It Catches	Execution Command
1. Formatting	`Prettier`	Spacing, quotes, line width (printWidth: 100), trailing commas, so all code looks the same across files and team members.	`npm run format:check` `npm run format`
2. Linting	`ESLint + typescript-eslint`	Bugs, implicit any, unused code/variables, bad import order, hook mistakes, and unhandled promises.	`npm run lint`
3. Type check	`tsc --noEmit (strict mode)`	Type errors, broken DTO contracts, undefined property accesses, missing function parameters, schema mismatches.	`npm run typecheck`
4. Tests	`Vitest + Testing Library`	Behaviour that breaks: tax calculation regressions, stock overselling, unauthorized tenant leaks, broken endpoint status codes.	`npm run test`
5. Gate	`Husky + lint-staged + CI`	Blocks bad code before commit and before merge. Automated pre-commit hook runs Layers 1–4 on staged files. Continuous Integration verifies full test matrix on PR.	Pre-commit hook & GitHub Actions CI

8. Quality Layer Execution & Verification Matrix

Mandatory Pre-Release Verification Protocol:

1. **Formatting Gate:** Run `npm run format:check` across `vyora-backend` and `vyora-frontend`. All source files must comply with shared Prettier rules.
2. **Linting Gate:** Run `npm run lint`. Zero ESLint warnings or errors permitted on production branches.
3. **Typecheck Gate:** Run `npm run typecheck` (which runs `tsc --noEmit`). Both projects must compile with 0 errors.
4. **Behavioral Test Suite:** Run `npm run test`. All NestJS controller, service, guard, and Vitest component tests must report 100% pass rate.
5. **Gatekeeper Certification:** Run `node scripts/ship-check.js`. Validates secret leakage prevention, UI audit compliance, and build artifacts before release.